Estimation Theory and Applications

Lecture 12

Prof. Jongeun Choi

Estimation Problem

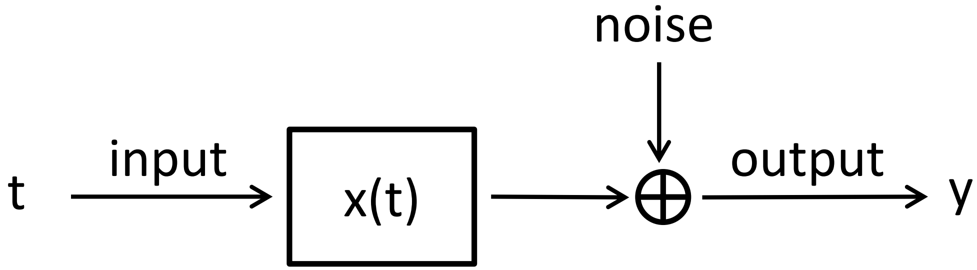
Bayesian Filtering

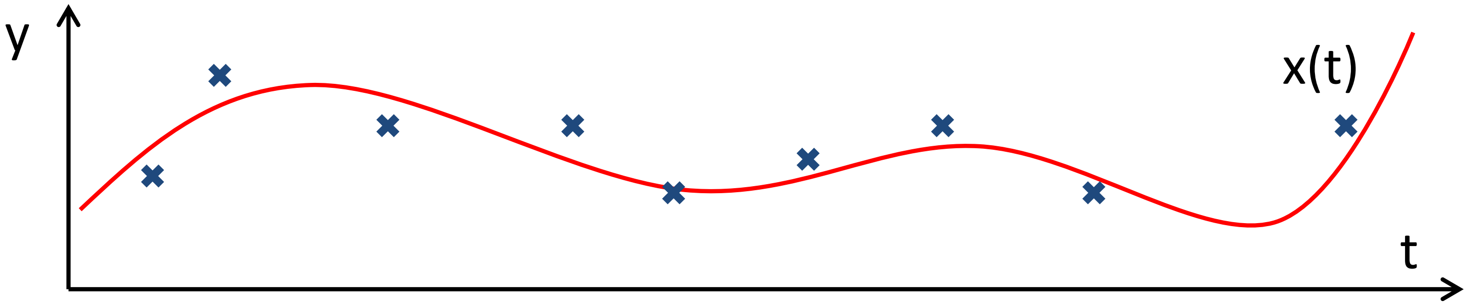
Kalman Filters – A Special Case of Bayesian Filtering

Other Filtering Algorithms: Extended Kalman Filters, Particle Filters

BAYESIAN FILTERING

Estimation Problem



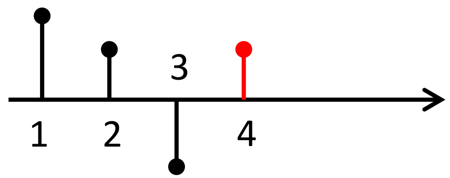


Estimation problem: Find x(t) from data $\{(t_n, y_n) : n=1, \ldots, N\}$

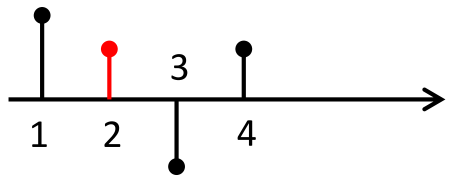
Estimation Problems

- Filtering
  - Estimate $x(t_N)$ from $\{(t_n, y_n) : n=1, \ldots, N\}$

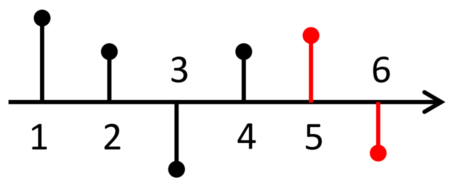
N=4



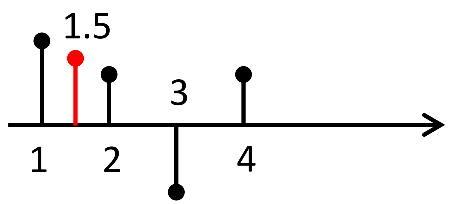
- Smoothing
  - Estimate $x(t_i)$ from $\{(t_n, y_n) : n=1, \ldots, N\}$, $1 \cdot i \cdot N$



- Prediction
  - Estimate $x(t_m)$ from $\{(t_n, y_n) : n=1, \ldots, N\}$, $m > N$



- Interpolation



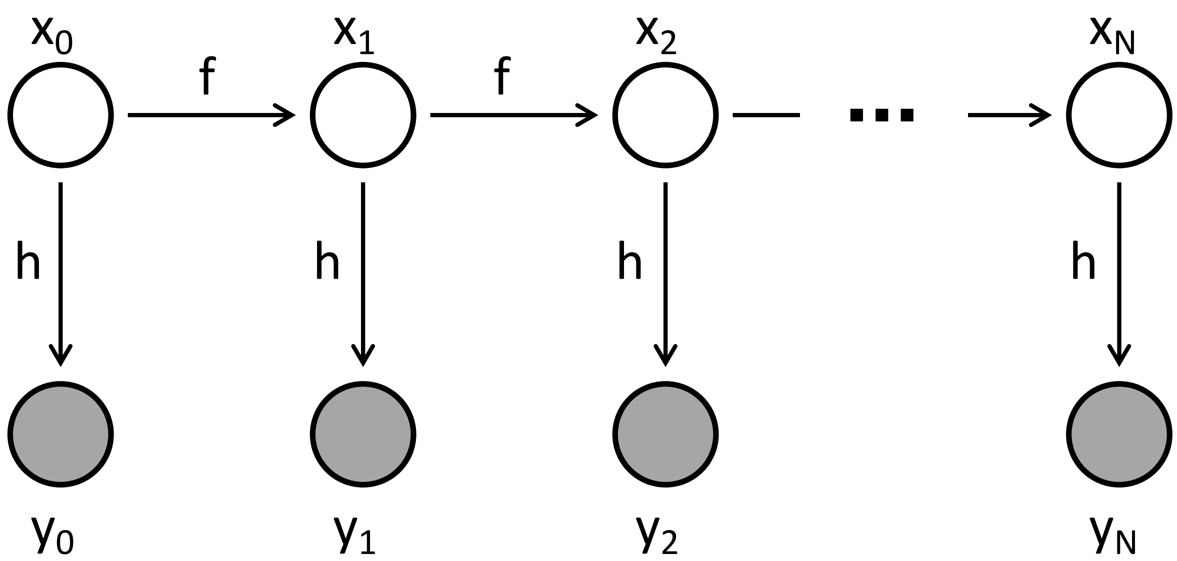
Dynamical Systems

Discrete-time dynamic and measurement model

$$\boxed{\begin{aligned} x_{t+1} &= f(x_t) + u_t \\ y_t &= h(x_t) + w_t \end{aligned}}$$

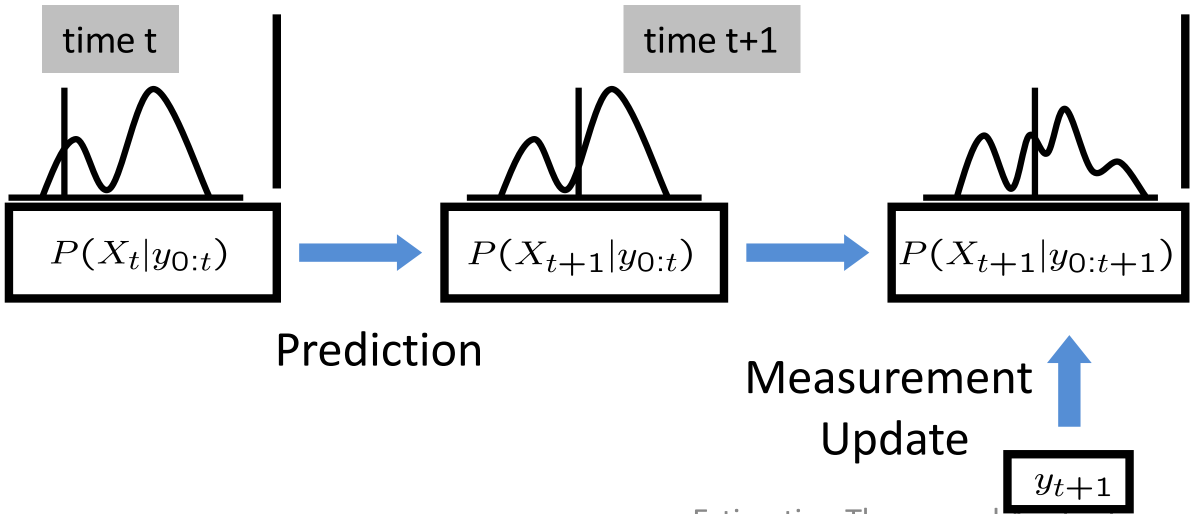
- x_t: state at time t
- y_t: observation at time t
- u_t, w_t: white noise processes

Graphical model:



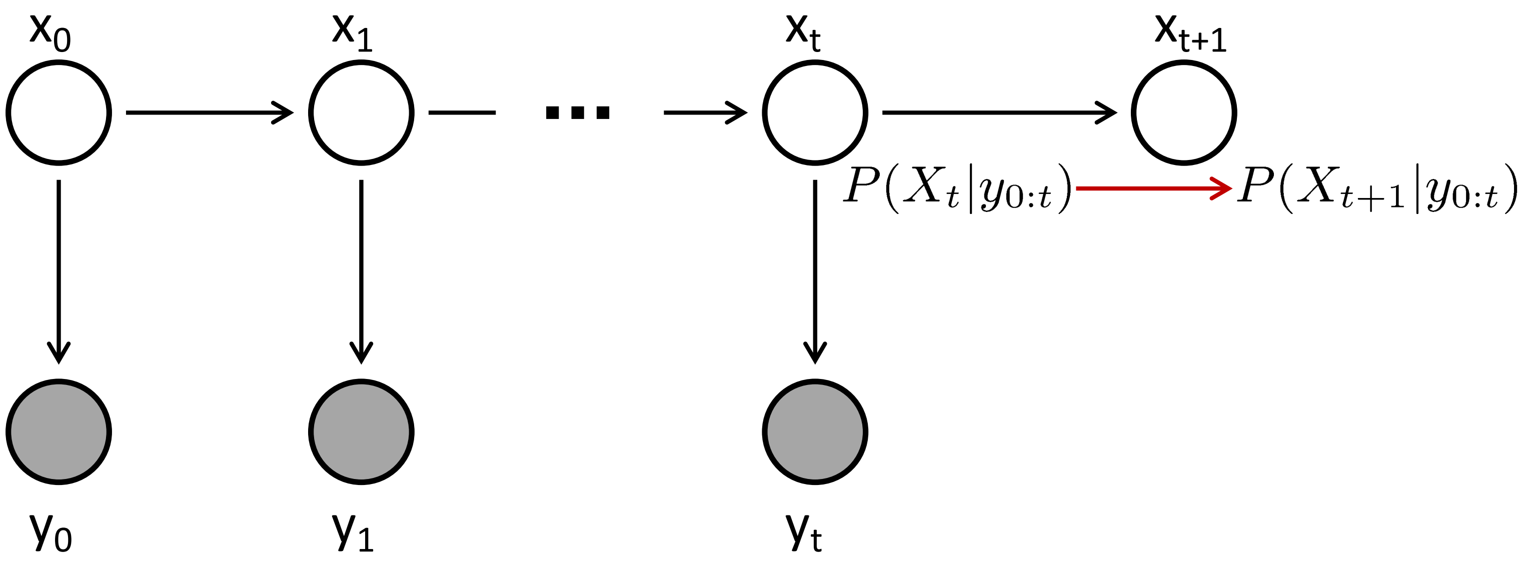
Bayesian Filtering

- Compute the posterior $P(X_t|y_{0:t})$ for each t, where $y_{0:t} = \{y_0, y_1, \ldots, y_t\}$.
- It is assumed that the prior $P(X_0)$ is known.
- Bayesian filtering is done in two steps: (1) prediction and (2) measurement update.
- Bayesian filtering algorithm
  1. Compute $P(X_{t+1}|y_{0:t})$ from $P(X_t|y_{0:t})$ (prediction)
  2. New observation y_{t+1}
  3. Compute $P(X_{t+1}|y_{0:t+1})$ (measurement update)
  4. Go to the next time



Examples of Bayesian filtering:
- Hidden Markov model,
- Kalman filters,
- Particle filters,
- …

Prediction



- $P(X_t|y_{0:t})$ is available from the previous time step.
- Prediction (or dynamic update):

$$\begin{aligned} P(X_{t+1}|y_{0:t}) &= \int P(X_{t+1}|x_t, y_{0:t})P(x_t|y_{0:t})dx_t \\ &= \int P(X_{t+1}|x_t)P(x_t|y_{0:t})dx_t \end{aligned}$$

Derivation

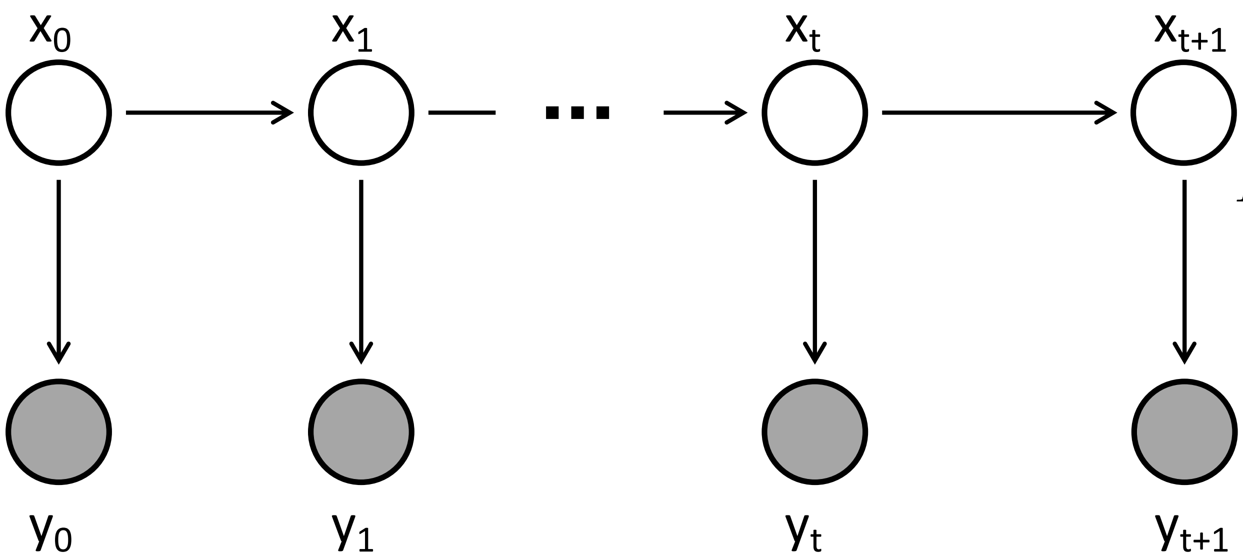
$$P(X|Y) = \frac{P(X,Y)}{P(Y)} = \frac{P(Y|X)P(X)}{P(Y)}$$

$$P(X|y,Y) = \frac{P(X,y|Y)}{P(y|Y)} = \frac{P(y|X,Y)P(X|Y)}{P(y|Y)}$$

$$P(X|y,Y) \propto P(y|X,Y)P(X|Y)$$

$$X = X_{t+1},\ y = y_{t+1},\ Y = y_{0:t}$$

Measurement Update



$P(X_{t+1}|y_{0:t}) \longrightarrow P(X_{t+1}|y_{0:t+1})$

- New measurement y_{t+1}.
- Measurement update (Bayes rule):

$$
\begin{aligned}
P(X_{t+1}|y_{0:t+1}) &= \frac{P(y_{t+1}|X_{t+1}, y_{0:t})P(X_{t+1}|y_{0:t})}{\int P(y_{t+1}|x_{t+1}, y_{0:t})P(x_{t+1}|y_{0:t})dx_{t+1}} \\
&= \frac{P(y_{t+1}|X_{t+1})P(X_{t+1}|y_{0:t})}{\int P(y_{t+1}|x_{t+1})P(x_{t+1}|y_{0:t})dx_{t+1}}
\end{aligned}
$$

A Special Case of Bayesian Filtering

KALMAN FILTERING

Rudolf Emil Kálmán

- **“Rudolf Emil Kálmán** (May 19, 1930 – July 2, 2016) was an Hungarian-American electrical engineer, mathematician, and inventor. He is most noted for his co-invention and development of the Kalman filter, a mathematical algorithm that is widely used in signal processing, control systems, and guidance, navigation and control. For this work, U.S. President Barack Obama awarded Kálmán the National Medal of Science on October 7, 2009.”

- “Kálmán's ideas on filtering were initially met with vast skepticism, so much so that he was forced to do the first publication of his results in mechanical engineering, rather than in electrical engineering or systems engineering. Kálmán had more success in presenting his ideas, however, while visiting Stanley F. Schmidt at the NASA Ames Research Center in 1960. This led to the use of Kálmán filters during the Apollo program, and furthermore, in the NASA Space Shuttle, in Navy submarines, and in unmanned aerospace vehicles and weapons, such as cruise missiles.”

-Wikipedia

Scholar

New results in linear filtering and prediction theory
RE Kalman, RS Bucy - 1961 - asmedigitalcollection.asme.org
A nonlinear differential equation of the Riccati type is derived for the covariance matrix of the optimal filtering error. The solution of this “variance equation” completely specifies the optimal filter for either finite or infinite smoothing intervals and stationary or nonstationary statistics. The variance equation is closely related to the Hamiltonian (canonical) differential equations of the calculus of variations. Analytic solutions are available in some cases. The significance of the variance equation is illustrated by examples which duplicate, simplify, or …
Cited by 7617 Related articles All 14 versions

- Kalman, R.E.; Bucy, R.S. (1961). "New Results in Linear Filtering and Prediction Theory" (PDF). *ASME Journal of Basic Engineering.*
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- Kalman, R. E. (1963). "Mathematical description of linear dynamical systems". *Journal of the Society for Industrial and Applied Mathematics.*
- Kalman, R. E.; Bertram, J. E. (1960). "Control system analysis and design Via the "second method" of Lyapunov: I — Continuous-time systems". *ASME Journal of Basic Engineering.*
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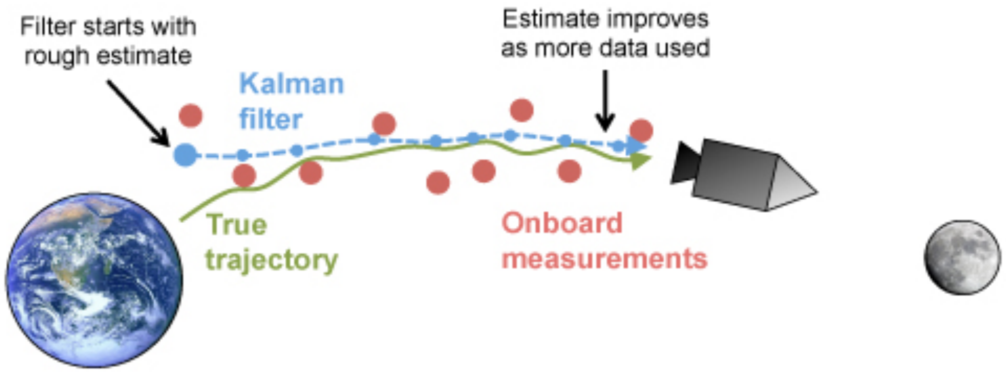
Application

- First application of Kalman filter was in the 1960s in the Apollo poject to estimate the trajectory of the spacecraft to the moon and back.



https://plus.maths.org/content/understanding-unseen By Adam Kucharski

Linear Gaussian Model

- Dynamic model

$$x_{t+1} = Ax_t + Gw_t$$

- Measurement model

$$y_t = Cx_t + v_t$$

- Initial state distribution: $x_0 \sim \mathcal{N}(\mu_0, \Sigma_0)$ (assume $\mu_0 = 0$ for now)
- Noises: $w_t \sim \mathcal{N}(0, Q)$, $v_t \sim \mathcal{N}(0, R)$. w_t, v_t, and x_0 are independent.

- Unconditional distribution of x_t (i.e., y_t are not observed)

$$\begin{aligned}
\mathbb{E}(x_t) &= 0 \\
\mathbf{var}(x_{t+1}) &= \Sigma_{t+1} = \mathbb{E}(x_{t+1}x_{t+1}^T) \\
&= \mathbb{E}\left((Ax_t + Gw_t)(Ax_t + Gw_t)^T\right) \\
&= A\mathbb{E}(x_t x_t^T)A^T + G\mathbb{E}(w_t w_t^T)G^T \\
&= A\Sigma_t A^T + GQG^T
\end{aligned}$$

Filtering

- Dynamic update

$$P(x_t|y_0,\dots,y_t) \to P(x_{t+1}|y_0,\dots,y_t) = \int P(x_{t+1}|x_t)P(x_t|y_0,\dots,y_t)dx_t$$

- Measurement update

$$P(x_{t+1}|y_0,\dots,y_t) \to P(x_{t+1}|y_0,\dots,y_t,y_{t+1}) \propto P(y_{t+1}|x_{t+1})P(x_{t+1}|y_0,\dots,y_t)$$

  - $P(x_{t+1}, y_{t+1}|y_0,\dots,y_t) \sim \mathcal{N}(\mu,\Sigma)$, where

$$\mu = \begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} \Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_{yy} \end{pmatrix}$$

Conditional PDF of Multivariate Gaussian

Theorem 10.2 (Conditional PDF of Multivariate Gaussian) *If* $\mathbf{x}$ *and* $\mathbf{y}$ *are jointly Gaussian, where* $\mathbf{x}$ *is* $k \times 1$ *and* $\mathbf{y}$ *is* $l \times 1$, *with mean vector* $[E(\mathbf{x})^T \; E(\mathbf{y})^T]^T$ *and partitioned covariance matrix*

$$\mathbf{C} = \begin{bmatrix} \mathbf{C}_{xx} & \mathbf{C}_{xy} \\ \mathbf{C}_{yx} & \mathbf{C}_{yy} \end{bmatrix} = \begin{bmatrix} k \times k & k \times l \\ l \times k & l \times l \end{bmatrix} \tag{10.23}$$

so that

$$p(\mathbf{x}, \mathbf{y}) = \frac{1}{(2\pi)^{\frac{k+l}{2}} \det^{\frac{1}{2}}(\mathbf{C})} \exp\left[-\frac{1}{2}\left(\begin{bmatrix} \mathbf{x} - E(\mathbf{x}) \\ \mathbf{y} - E(\mathbf{y}) \end{bmatrix}\right)^T \mathbf{C}^{-1} \left(\begin{bmatrix} \mathbf{x} - E(\mathbf{x}) \\ \mathbf{y} - E(\mathbf{y}) \end{bmatrix}\right)\right],$$

then the conditional PDF $p(\mathbf{y}|\mathbf{x})$ *is also Gaussian and*

$$E(\mathbf{y}|\mathbf{x}) = E(\mathbf{y}) + \mathbf{C}_{yx}\mathbf{C}_{xx}^{-1}(\mathbf{x} - E(\mathbf{x})) \tag{10.24}$$

$$\mathbf{C}_{y|x} = \mathbf{C}_{yy} - \mathbf{C}_{yx}\mathbf{C}_{xx}^{-1}\mathbf{C}_{xy}. \tag{10.25}$$

Filtering

- Dynamic update

$$P(x_t|y_0,\ldots,y_t) \to P(x_{t+1}|y_0,\ldots,y_t) = \int P(x_{t+1}|x_t)P(x_t|y_0,\ldots,y_t)dx_t$$

- Measurement update

$$P(x_{t+1}|y_0,\ldots,y_t) \to P(x_{t+1}|y_0,\ldots,y_t,y_{t+1}) \propto P(y_{t+1}|x_{t+1})P(x_{t+1}|y_0,\ldots,y_t)$$

  - $P(x_{t+1},y_{t+1}|y_0,\ldots,y_t) \sim \mathcal{N}(\mu,\Sigma)$, where

$$\mu = \begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} \Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_{yy} \end{pmatrix}$$

  - $P(x_{t+1}|y_0,\ldots,y_t,y_{t+1}) \sim \mathcal{N}(\mu_{x|y},\Sigma_{x|y})$, where

$$\begin{aligned} \mu_{x|y} &= \mu_x + \Sigma_{xy}\Sigma yy^{-1}(y-\mu_y) \\ \Sigma_{x|y} &= \Sigma_{xx} - \Sigma_{xy}\Sigma_{yy}^{-1}\Sigma_{yx} \end{aligned}$$

Dynamic Update

$$\begin{aligned}\hat{x}_{t|t} &:= \mathbb{E}(x_t|y_0,\ldots,y_t)\\ P_{t|t} &= \mathbb{E}(\tilde{x}_{t|t}\tilde{x}_{t|t}^T|y_0,\ldots,y_t)\\ \tilde{x}_{t|t} &= x_t - \hat{x}_{t|t}\end{aligned} \quad \Longrightarrow \quad \begin{aligned}\hat{x}_{t+1|t} &:= \mathbb{E}(x_{t+1}|y_0,\ldots,y_t)\\ P_{t+1|t} &= \mathbb{E}\left(\tilde{x}_{t+1|t}\tilde{x}_{t+1|t}^T|y_0,\ldots,y_t\right)\\ \tilde{x}_{t+1|t} &= x_{t+1} - \hat{x}_{t+1|t}\end{aligned}$$

$$\begin{aligned}\hat{x}_{t+1|t} &= \mathbb{E}(x_{t+1}|y_0,\ldots,y_t)\\ &= \mathbb{E}(Ax_t + Gw_t|y_0,\ldots,y_t)\\ &= A\hat{x}_{t|t}\end{aligned}$$

$$x_{t+1} = Ax_t + Gw_t$$

$$\begin{aligned}P_{t+1|t} &= \mathbb{E}\left[\tilde{x}_{t+1|t}\tilde{x}_{t+1|t}^T|y_0,\ldots,y_t\right]\\ &= \mathbb{E}\left[(Ax_t + Gw_t - A\hat{x}_{t|t})(Ax_t + Gw_t - A\hat{x}_{t|t})^T|y_0,\ldots,y_t\right]\\ &= \mathbb{E}\left[(A(x_t - \hat{x}_{t|t}) + Gw_t)(A(x_t - \hat{x}_{t|t}) + Gw_t)^T|y_0,\ldots,y_t\right]\\ &= AP_{t|t}A^T + GQG^T\end{aligned}$$

Measurement Update (1)

$$\mu_x \quad \hat{x}_{t+1|t} := \mathbb{E}(x_{t+1}|y_0, \ldots, y_t) \qquad \hat{x}_{t+1|t+1} := \mathbb{E}(x_{t+1}|y_0, \ldots, y_{t+1})$$

$$\Sigma_{xx} \quad P_{t+1|t} = \mathbb{E}\left(\tilde{x}_{t+1|t}\tilde{x}_{t+1|t}^T|y_0, \ldots, y_t\right) \Longrightarrow P_{t+1|t+1} = \mathbb{E}(\tilde{x}_{t+1|t+1}\tilde{x}_{t+1|t+1}^T|y_0, \ldots, y_{t+1})$$

$$\tilde{x}_{t+1|t} = x_{t+1} - \hat{x}_{t+1|t} \qquad \tilde{x}_{t+1|t+1} = x_{t+1} - \hat{x}_{t+1|t+1}$$

Distribution of y_{t+1} given $y_0, \ldots, y_t$:

$$\mu_y \quad \begin{aligned} \hat{y}_{t+1|t} &:= \mathbb{E}(y_{t+1}|y_0, \ldots, y_t) \\ &= \mathbb{E}(Cx_{t+1} + v_{t+1}|y_0, \ldots, y_t) = C\hat{x}_{t+1|t} \end{aligned}$$

$$\Sigma_{yy} \quad \begin{aligned} &\mathbb{E}\left[(y_{t+1} - \hat{y}_{t+1|t})(y_{t+1} - \hat{y}_{t+1|t})^T|y_0, \ldots, y_t\right] \\ &= \mathbb{E}\left[(Cx_{t+1} + v_{t+1} - C\hat{x}_{t+1|t})(Cx_{t+1} + v_{t+1} - C\hat{x}_{t+1|t})^T|y_0, \ldots, y_t\right] \\ &= \mathbb{E}\left[(C(x_{t+1} - \hat{x}_{t+1|t}) + v_{t+1})(C(x_{t+1} - \hat{x}_{t+1|t}) + v_{t+1})^T|y_0, \ldots, y_t\right] \\ &= CP_{t+1|t}C^T + R \end{aligned}$$

$$\Sigma_{yx} \quad \begin{aligned} &\mathbb{E}\left[(y_{t+1} - \hat{y}_{t+1|t})(x_{t+1} - \hat{x}_{t+1|t})^T|y_0, \ldots, y_t\right] \\ &= \mathbb{E}\left[(Cx_{t+1} + v_{t+1} - C\hat{x}_{t+1|t})(x_{t+1} - \hat{x}_{t+1|t})^T|y_0, \ldots, y_t\right] \\ &= CP_{t+1|t} \end{aligned}$$

Measurement Update (2)

x_{t+1} and y_{t+1} have the conditional joint multivariate Gaussian distribution

$$P(x_{t+1}, y_{t+1} | y_0, \ldots, y_t) \sim \mathcal{N}(\mu, \Sigma),$$

where

$$\mu = \begin{pmatrix} \hat{x}_{t+1|t} \\ C\hat{x}_{t+1|t} \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} P_{t+1|t} & P_{t+1|t}C^T \\ CP_{t+1|t} & CP_{t+1|t}C^T + R \end{pmatrix}$$

- $P(x_{t+1}, y_{t+1} | y_0, \ldots, y_t) \sim \mathcal{N}(\mu, \Sigma)$, where

$$\mu = \begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} \Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_{yy} \end{pmatrix}$$

- $P(x_{t+1} | y_0, \ldots, y_t, y_{t+1}) \sim \mathcal{N}(\mu_{x|y}, \Sigma_{x|y})$, where

$$\begin{aligned} \mu_{x|y} &= \mu_x + \Sigma_{xy}\Sigma yy^{-1}(y - \mu_y) \\ \Sigma_{x|y} &= \Sigma_{xx} - \Sigma_{xy}\Sigma_{yy}^{-1}\Sigma_{yx} \end{aligned}$$

$P(x_{t+1} | y_0, \ldots, y_t, y_{t+1}) \sim \mathcal{N}(\hat{x}_{t+1|t+1}, P_{t+1|t+1})$, where

$$\begin{aligned} \hat{x}_{t+1|t+1} &= \hat{x}_{t+1|t} + P_{t+1|t}C^T(CP_{t+1|t}C^T + R)^{-1}(y_{t+1} - C\hat{x}_{t+1|t}) \\ P_{t+1|t+1} &= P_{t+1|t} - P_{t+1|t}C^T(CP_{t+1|t}C^T + R)^{-1}CP_{t+1|t} \end{aligned}$$

Kalman Filter

- Dynamic update

$$\begin{aligned} \hat{x}_{t+1|t} &= A\hat{x}_{t|t} \\ P_{t+1|t} &= AP_{t|t}A^T + GQG^T \end{aligned}$$

- Measurement update

$$\begin{aligned} \hat{x}_{t+1|t+1} &= \hat{x}_{t+1|t} + K_{t+1}(y_{t+1} - C\hat{x}_{t+1|t}) \\ P_{t+1|t+1} &= P_{t+1|t} - K_{t+1}CP_{t+1|t} \end{aligned}$$

- Kalman gain

$$K_{t+1} = P_{t+1|t}C^T(CP_{t+1|t}C^T + R)^{-1}$$

- Dynamic model: $x_{t+1} = Ax_t + Gw_t$
- Measurement model: $y_t = Cx_t + v_t$
- Initial state distribution: $x_0 \sim \mathcal{N}(\mu_0, \Sigma_0)$
- Noises: $w_t \sim \mathcal{N}(0, Q)$, $v_t \sim \mathcal{N}(0, R)$. w_t, v_t, and x_t are independent.

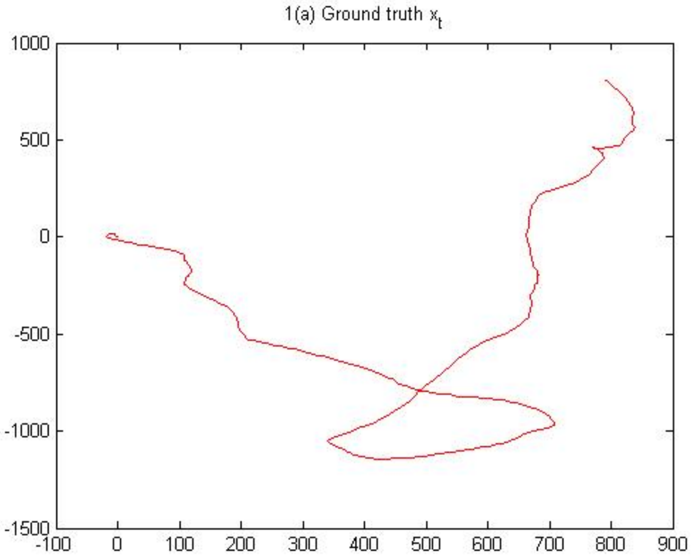
Example (1)

A particle moving in a plane under random forces and damping.

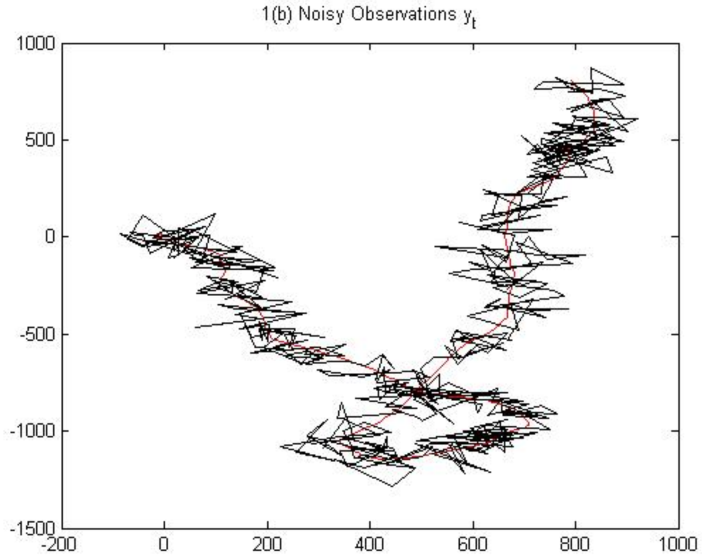
- State of the particle: $x = [x^1, \dot{x}^1, x^2, \dot{x}^2]^T$
- (x^1, x^2): position of the particle; $(\dot{x}^1, \dot{x}^2)$: velocity of the particle
- $x^i_{t+1} = x^i_t + \dot{x}^i_t$
- $\dot{x}^i_{t+1} = 0.98\dot{x}^i_t + w^i_t$
- Dynamic model: $x_{t+1} = Ax_t + Gw_t$
- Measurement model: $y_t = Cx_t + v_t$

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0.98 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0.98 \end{bmatrix} \quad G = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

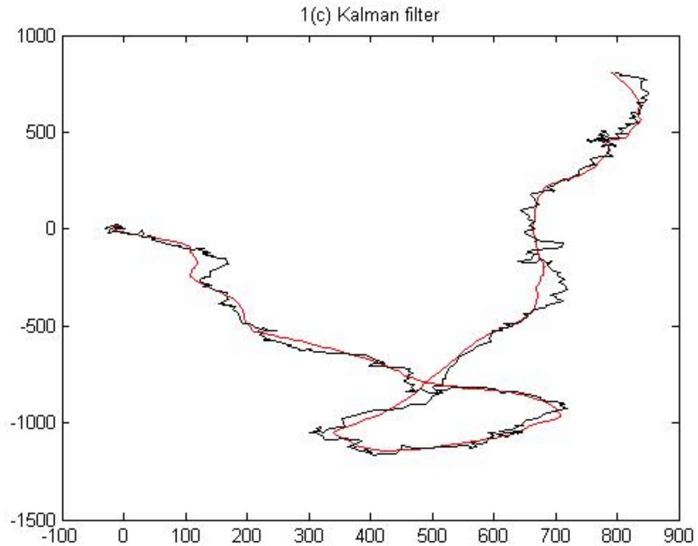
Example (2)



Ground Truth



Noisy Measurements



Kalman filter

Extended Kalman Filters

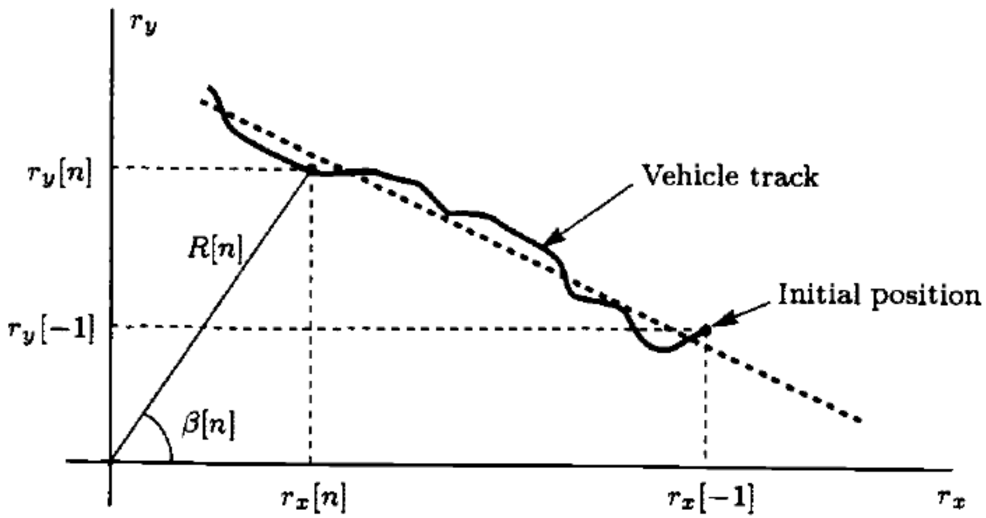
Particle Filters

OTHER FILTERING ALGORITHMS

Extended Kalman Filters (1)

- When the dynamic or measurement is nonlinear, previous results cannot be used.
- But we can still apply the previous results by <u>linearizing</u> the nonlinear functions.

Vehicle tracking example



Vehicle's position: (r_x, r_y)

Nonlinear measurement model

$$\begin{aligned} \hat{R}[n] &= R[n] + w_R[n] \\ \hat{\beta}[n] &= \beta[n] + w_\beta[n] \end{aligned}$$

$$R[n] = \sqrt{r_x^2[n] + r_y^2[n]}$$ Range

$$\beta[n] = \arctan \frac{r_y[n]}{r_x[n]}.$$ Bearing (angle)

Source: Steven M. Kay, "Fundamentals of Statistical Signal Processing, Volume I: Estimation Theory", Prentice Hall, 1993.

Extended Kalman Filters (2)

Original nonlinear system:

$$\begin{aligned}\mathbf{s}[n] &= \mathbf{a}(\mathbf{s}[n-1]) + \mathbf{B}\mathbf{u}[n] \\ \mathbf{x}[n] &= \mathbf{h}(\mathbf{s}[n]) + \mathbf{w}[n]\end{aligned}$$

a(.) and h(.) are nonlinear vector valued functions

$\mathbf{u}[n] \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}) \quad \mathbf{w}[n] \sim \mathcal{N}(\mathbf{0}, \mathbf{C}[n])$

First-order Taylor expansion:

$$\begin{aligned}\mathbf{a}(\mathbf{s}[n-1]) &\approx \mathbf{a}(\hat{\mathbf{s}}[n-1|n-1]) \\ &\quad + \left.\frac{\partial \mathbf{a}}{\partial \mathbf{s}[n-1]}\right|_{\mathbf{s}[n-1]=\hat{\mathbf{s}}[n-1|n-1]} (\mathbf{s}[n-1] - \hat{\mathbf{s}}[n-1|n-1]) \\ \mathbf{h}(\mathbf{s}[n]) &\approx \mathbf{h}(\hat{\mathbf{s}}[n|n-1]) + \left.\frac{\partial \mathbf{h}}{\partial \mathbf{s}[n]}\right|_{\mathbf{s}[n]=\hat{\mathbf{s}}[n|n-1]} (\mathbf{s}[n] - \hat{\mathbf{s}}[n|n-1]) .\end{aligned}$$

Let

$$\begin{aligned}\mathbf{A}[n-1] &= \left.\frac{\partial \mathbf{a}}{\partial \mathbf{s}[n-1]}\right|_{\mathbf{s}[n-1]=\hat{\mathbf{s}}[n-1|n-1]} \\ \mathbf{H}[n] &= \left.\frac{\partial \mathbf{h}}{\partial \mathbf{s}[n]}\right|_{\mathbf{s}[n]=\hat{\mathbf{s}}[n|n-1]}\end{aligned}$$

Linearized dynamic and measurement models

$$\begin{aligned}\mathbf{s}[n] &= \mathbf{A}[n-1]\mathbf{s}[n-1] + \mathbf{B}\mathbf{u}[n] + (\mathbf{a}(\hat{\mathbf{s}}[n-1|n-1]) - \mathbf{A}[n-1]\hat{\mathbf{s}}[n-1|n-1]) \\ \mathbf{x}[n] &= \mathbf{H}[n]\mathbf{s}[n] + \mathbf{w}[n] + (\mathbf{h}(\hat{\mathbf{s}}[n|n-1]) - \mathbf{H}[n]\hat{\mathbf{s}}[n|n-1]) .\end{aligned}$$

Extended Kalman Filters (3)

Prediction:

$$\hat{\mathbf{s}}[n|n-1] = \mathbf{a}(\hat{\mathbf{s}}[n-1|n-1]). \tag{13.67}$$

Minimum Prediction MSE Matrix ($p \times p$):

$$\mathbf{M}[n|n-1] = \mathbf{A}[n-1]\mathbf{M}[n-1|n-1]\mathbf{A}^T[n-1] + \mathbf{BQB}^T. \tag{13.68}$$

Kalman Gain Matrix ($p \times M$):

$$\mathbf{K}[n] = \mathbf{M}[n|n-1]\mathbf{H}^T[n]\left(\mathbf{C}[n] + \mathbf{H}[n]\mathbf{M}[n|n-1]\mathbf{H}^T[n]\right)^{-1}. \tag{13.69}$$

Correction:

$$\hat{\mathbf{s}}[n|n] = \hat{\mathbf{s}}[n|n-1] + \mathbf{K}[n](\mathbf{x}[n] - \mathbf{h}(\hat{\mathbf{s}}[n|n-1])). \tag{13.70}$$

Minimum MSE Matrix ($p \times p$):

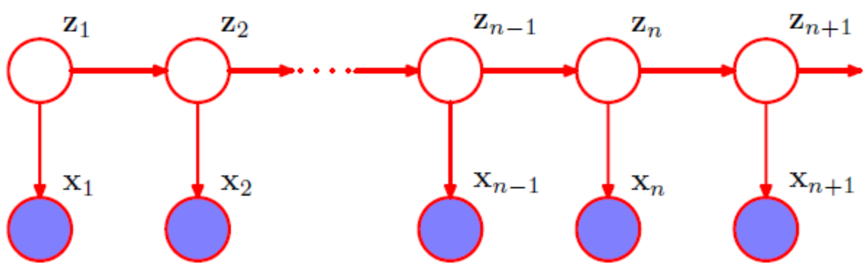
$$\mathbf{M}[n|n] = (\mathbf{I} - \mathbf{K}[n]\mathbf{H}[n])\mathbf{M}[n|n-1] \tag{13.71}$$

where

$$\begin{aligned}
\mathbf{A}[n-1] &= \left.\frac{\partial \mathbf{a}}{\partial \mathbf{s}[n-1]}\right|_{\mathbf{s}[n-1]=\hat{\mathbf{s}}[n-1|n-1]} \\
\mathbf{H}[n] &= \left.\frac{\partial \mathbf{h}}{\partial \mathbf{s}[n]}\right|_{\mathbf{s}[n]=\hat{\mathbf{s}}[n|n-1]}.
\end{aligned}$$

Particle Filters (1)

- For dynamical systems which is not linear-Gaussian, a particle filter uses a sampling method to approximate the posterior distribution.
- Particle filtering is also known as *sequential Monte Carlo* or a *condensation* algorithm in computer vision.



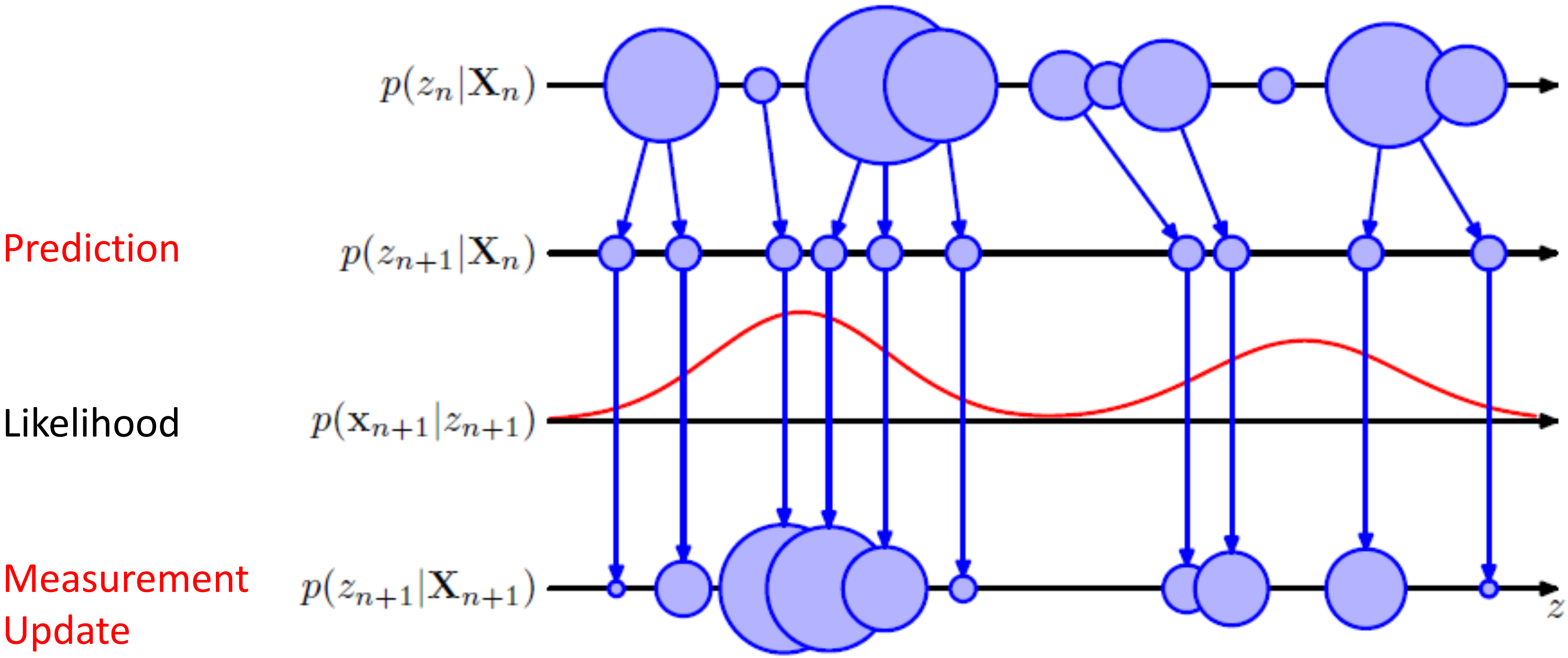
$$\begin{aligned}
\mathbb{E}[f(\mathbf{z}_n)] &= \int f(\mathbf{z}_n) p(\mathbf{z}_n|\mathbf{X}_n)\,\mathrm{d}\mathbf{z}_n \\
&= \int f(\mathbf{z}_n) p(\mathbf{z}_n|\mathbf{x}_n, \mathbf{X}_{n-1})\,\mathrm{d}\mathbf{z}_n \\
&= \frac{\int f(\mathbf{z}_n) p(\mathbf{x}_n|\mathbf{z}_n) p(\mathbf{z}_n|\mathbf{X}_{n-1})\,\mathrm{d}\mathbf{z}_n}{\int p(\mathbf{x}_n|\mathbf{z}_n) p(\mathbf{z}_n|\mathbf{X}_{n-1})\,\mathrm{d}\mathbf{z}_n} \\
&\simeq \sum_{l=1}^{L} w_n^{(l)} f(\mathbf{z}_n^{(l)})
\end{aligned}$$

- z_n: state at time n
- $\mathbf{X}_n = (x_1, \ldots, x_n)$: measurements
- $z_n^{(l)}$: a sample from $p(z_n|\mathbf{X}_{n-1})$
- $w_n^{(l)}$: sampling weight

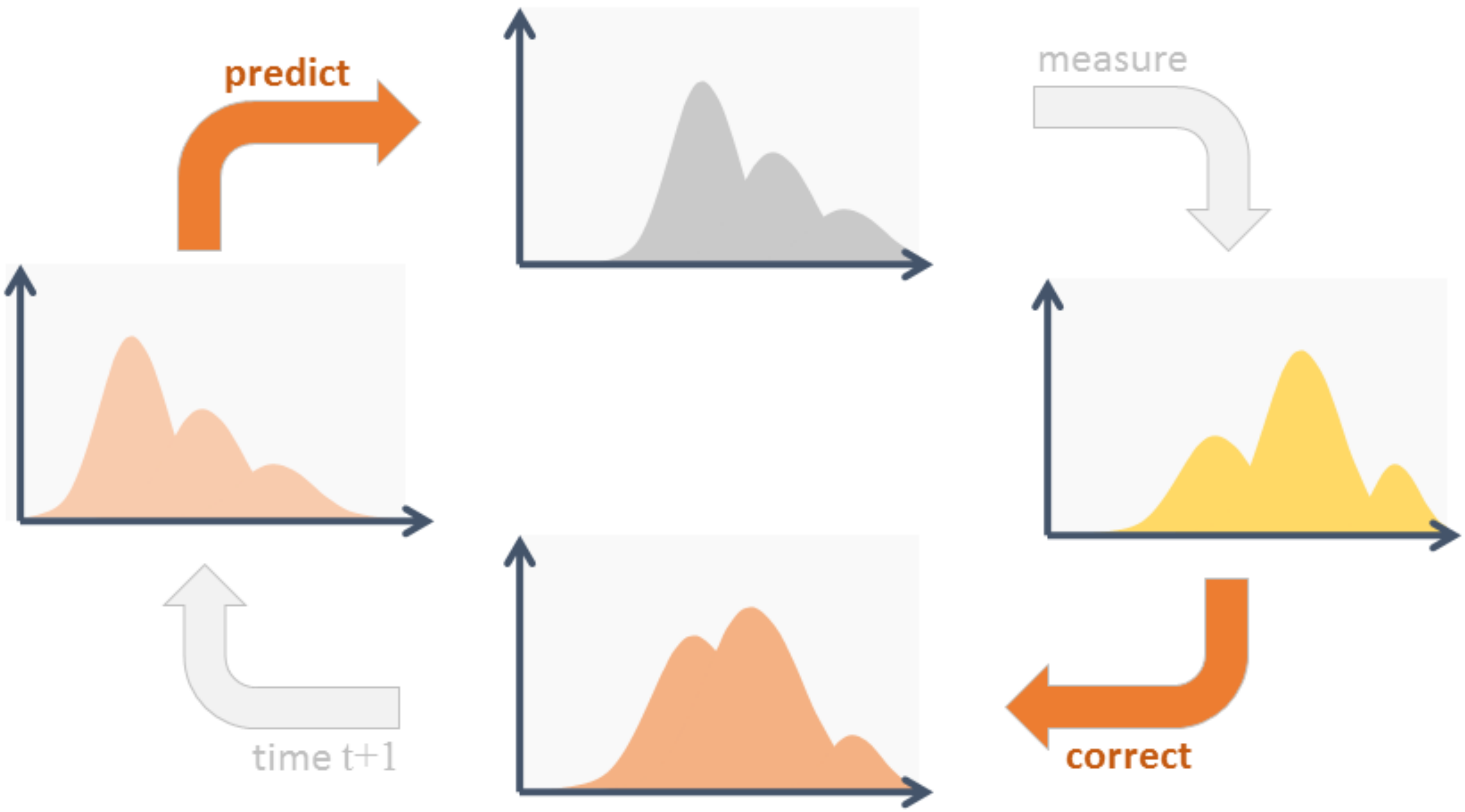
$$w_n^{(l)} = \frac{p(\mathbf{x}_n|\mathbf{z}_n^{(l)})}{\sum_{m=1}^{L} p(\mathbf{x}_n|\mathbf{z}_n^{(m)})}$$

Source: Christopher M. Bishop, "Pattern Recognition and Machine Learning", Springer, 2007.

Particle Filters (2)

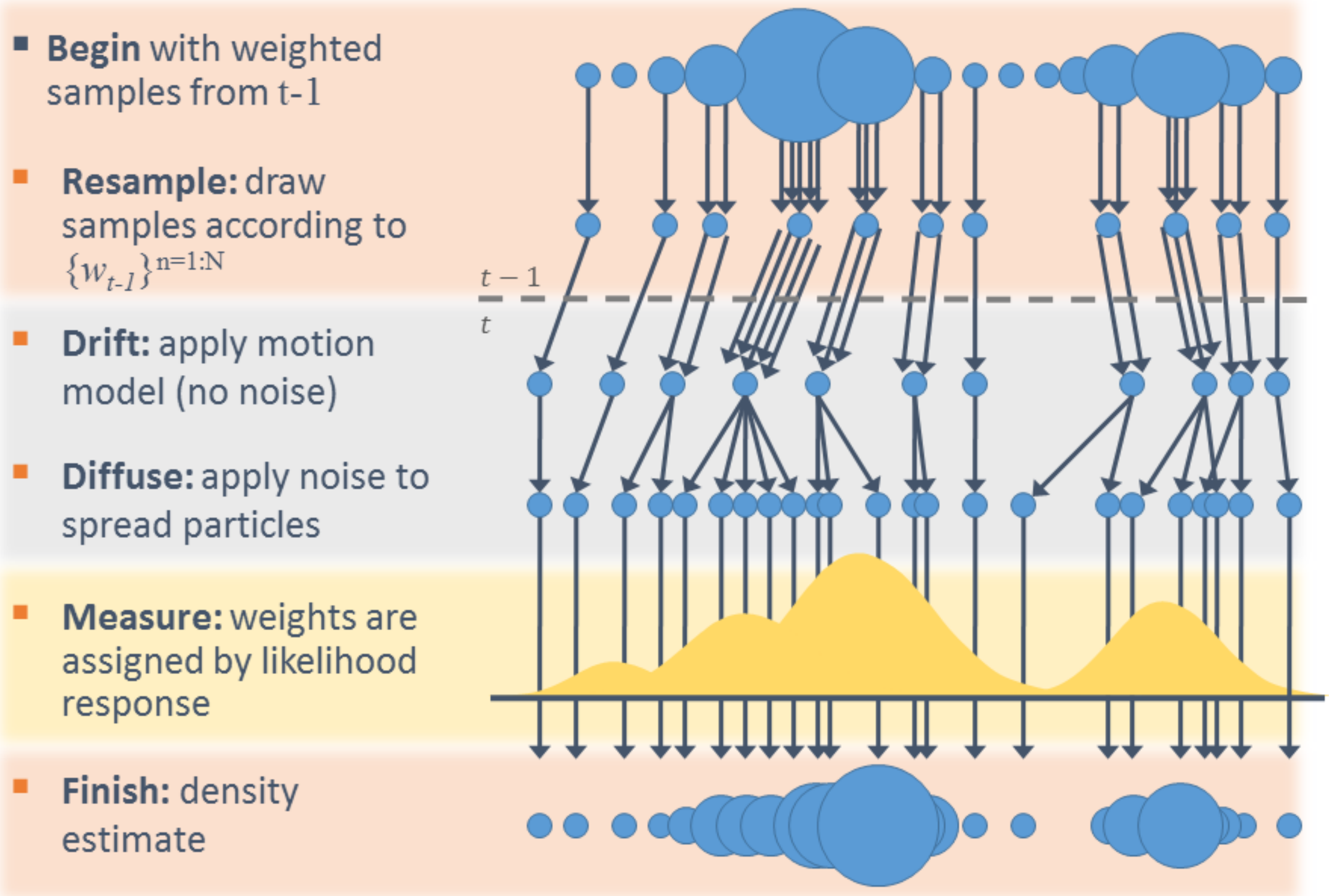


Cycle of particle filtering



Pictures from https://www.codeproject.com/Articles/865934/Object-Tracking-Particle-Filter-with-Ease#opticalFlow

Particle filtering steps



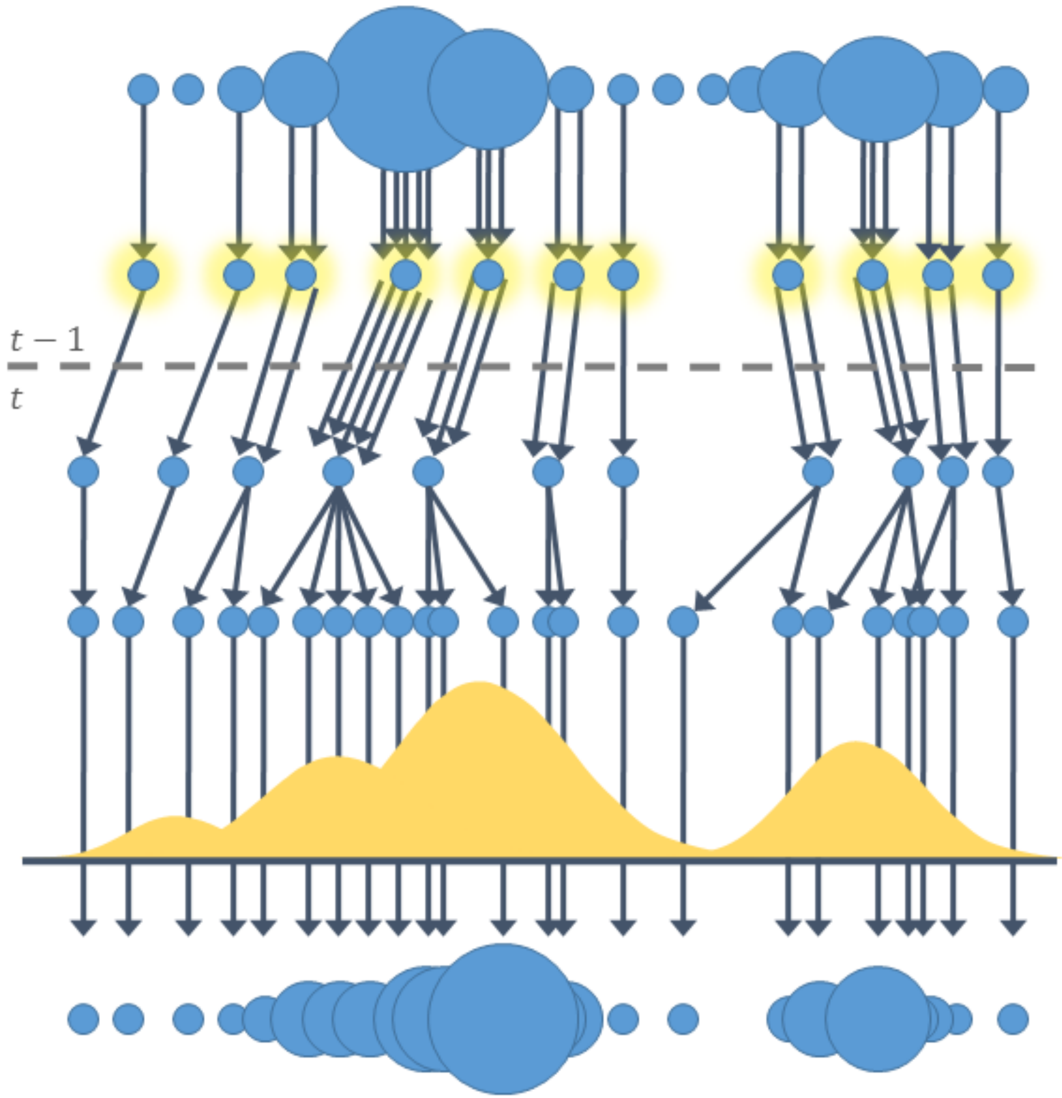
Pictures from https://www.codeproject.com/Articles/865934/Object-Tracking-Particle-Filter-with-Ease#opticalFlow

Prediction step: Re-sampling

- **Resample:** draw samples according to $\{w_{t-1}\}^{n=1:N}$

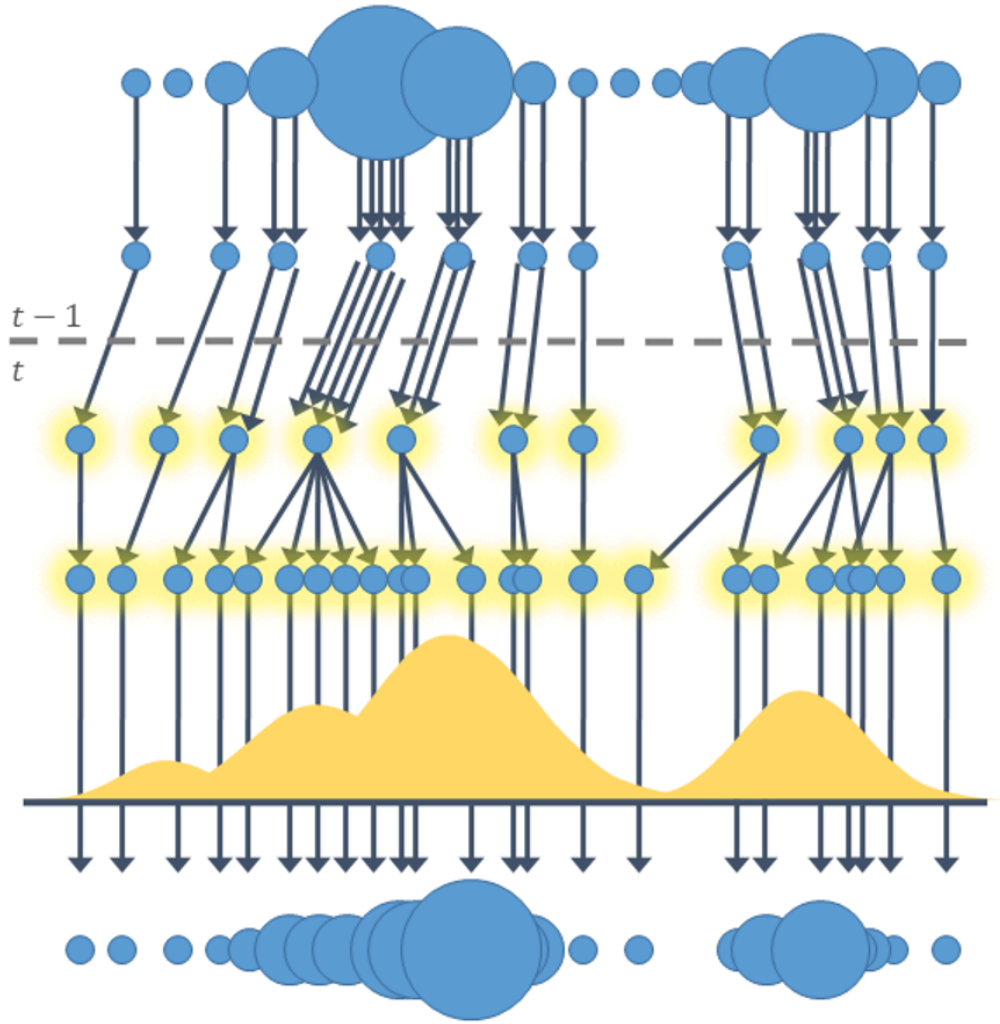
N new samples are drawn from the previous set **with replacement.**

New samples are assigned **uniform weights.**



Prediction step

- Prediction: Apply the motion model $p(z_{n+1}|X_n)$
- $z_{n+1}=f(z_n,v_n)$
- Drift: apply motion model (no noise)
- Diffuse: apply noise to spread particles



Update (Correction) step

- Obtain an observation x_{n+1} to estimate z_{n+1}
- Evaluate likelihood that Z_{n+1} give rise to x_{n+1} using observation model
  - $P(X_{n+1} | z_{n+1})$
- Measurement update: weights are proportional to the observation likelihood
  - $P(z_{n+1} | X_{n+1})$